

File permissions in Linux

Project description

In this case, I must examine and manage the file permissions in the `/home/researcher2/projects` directory for the user researcher2`.`

The user `researcher2` belongs to the research_team` group.`

I must verify the permissions for all files in the directory—including hidden files—to ensure they align with the appropriate level of authorization. If they do not, you must change the permissions.

Here are the steps I will follow: First, I'll check the user and group permissions for all files in the `projects` directory. Next, I'll identify any files with permissions and if I need to modify them as needed. Finally, I will check the permissions for the /home/researcher2/drafts` directory and modify them to remove unauthorized access.`

Check file and directory details

```
researcher2@790206c49713:~$ pwd
/home/researcher2
researcher2@790206c49713:~$ ls
projects
researcher2@790206c49713:~$ cd /home/researcher2/projects
researcher2@790206c49713:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Jul 23 02:59 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jul 23 02:59 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 23 02:59 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 23 02:59 project_t.txt
researcher2@790206c49713:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 02:59 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 03:12 ..
-rw--w---- 1 researcher2 research_team 46 Jul 23 02:59 .project_x.tx
t
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Jul 23 02:59 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jul 23 02:59 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 23 02:59 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 23 02:59 project_t.txt
researcher2@790206c49713:~/projects$ cd /home/researcher2/projects/dra
fts
researcher2@790206c49713:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 02:59 ..
researcher2@790206c49713:~/projects/drafts$ █
```

I used the `ls` command with the `-la` option to display a detailed listing of the file contents that also returned hidden files. The output of my command indicates that there is one directory named `drafts`, one hidden file named `.project_x.txt`, and five other project files. The 10-character string in the first column represents the permissions set on each file or directory.

Describe the permissions string

In the `/home/researcher2/projects` directory, there are five files with the following names and permissions:

- `project_k.txt`
 - User = read, write,
 - Group = read, write
 - Other = read, write
- `project_m.txt`

- User = read, write
 - Group = read
 - Other = none
- `project_r.txt`
 - User = read, write
 - Group = read, write
 - Other = read
- `project_t.txt`
 - User = read, write
 - Group = read, write
 - Other = read
- `.project_x.txt`
 - User = read, write
 - Group = write
 - Other = none

There is also one subdirectory inside the `projects` directory named `drafts`. The permissions on `drafts` are:

- User = read, write, execute
- Group = execute
- Other = none

Change file permissions

I determined that other shouldn't have write access to any of their files. To comply with this, I referred to the file permissions that I previously returned. I determined `project_k.txt` must have the write access removed for other.

Now, I update/change the next permissions:

- `project_k.txt`
 - User = read, write,
 - Group = read, write
 - Other = read
- `project_m.txt`
 - User = read, write
 - Group = none
 - Other = none

The following code demonstrates how I used Linux commands to do this:

```

researcher2@790206c49713:~/projects$ chmod o-w project_k.txt
researcher2@790206c49713:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jul 23 02:59 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_t.txt
researcher2@790206c49713:~/projects$ chmod g-r project_m.txt
researcher2@790206c49713:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_k.txt
-rw----- 1 researcher2 research_team  46 Jul 23 02:59 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_t.txt
researcher2@790206c49713:~/projects$ █

```

The `chmod` command changes the permissions on files and directories. The first argument indicates what permissions should be changed, and the second argument specifies the file or directory. In this example, I removed write permissions from other for the `project_k.txt` file. After this, I used `ls -la` to review the updates I made.

Change file permissions on a hidden file

I do not want anyone to have write access to this project, but the user and group should have read access.

The following code demonstrates how I used Linux commands to change the permissions:

```

researcher2@790206c49713:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@790206c49713:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 02:59 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 03:12 ..
-r--r----- 1 researcher2 research_team  46 Jul 23 02:59 .project_x.tx
t
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_k.txt
-rw----- 1 researcher2 research_team  46 Jul 23 02:59 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jul 23 02:59 project_t.txt
researcher2@790206c49713:~/projects$

```

I know `.project_x.txt` is a hidden file because it starts with a period (`.`). In this example, I removed write permissions from the user and group, and added read permissions to the group. I removed write permissions from the user with `u-w`. Then, I removed write permissions from the group with `g-w`, and added read permissions to the group with `g+r`.

Change directory permissions

I modified a directory's permissions, but I made an error because I forgot to include the `chmod` command in the command line. I noticed the mistake, corrected it, and learned the importance of carefully reviewing commands before executing them.

```
researcher2@790206c49713:~/projects$ cd /home/researcher2/projects/drafts
researcher2@790206c49713:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Jul 23 02:59 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 02:59 ..
researcher2@790206c49713:~/projects/drafts$ cd /home/researcher2/projects/drafts
researcher2@790206c49713:~/projects$ g-x drafts
-bash: g-x: command not found
researcher2@790206c49713:~/projects$ chmod g-x drafts
researcher2@790206c49713:~/projects$ cd /home/researcher2/projects/drafts
researcher2@790206c49713:~/projects/drafts$ ls -la
total 8
drwx----- 2 researcher2 research_team 4096 Jul 23 02:59 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 23 02:59 ..
researcher2@790206c49713:~/projects/drafts$
```

Summary

I really enjoy exploring ways to analyze file and directory permissions, as well as learning how to modify and manage them. I think it's a great path toward improving my understanding of access control and authorization management in Linux. The first step in this was using `ls -la` to check the permissions for the directory. This informed my decisions in the following steps. I then used the `chmod` command multiple times to change the permissions on files and directories.

